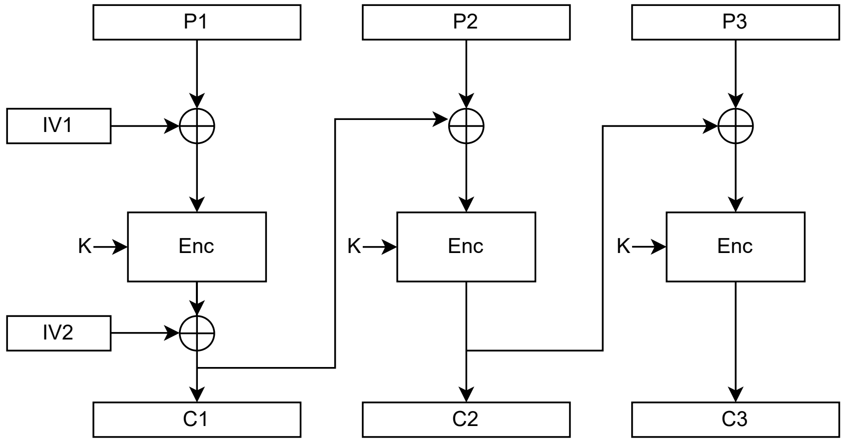

Consider the above block cipher encryption diagram. Write down the encryption equations for C_1, C_2, and C_3.

$$\begin{aligned} C_1 &= E_K(P_1 \oplus IV_1) \oplus IV_2 \\ C_2 &= E_K(P_2 \oplus C_1) \\ C_3 &= E_K(P_3 \oplus C_2) \end{aligned}$$

Write down the decryption equations for P_1, P_2, and P_3.

$$\begin{aligned} P_1 &= D_K(C_1 \oplus IV_2) \oplus IV_1 \\ P_2 &= D_K(C_2) \oplus C_1 \\ P_3 &= D_K(C_3) \oplus C_2 \end{aligned}$$

You are Mallory. You know the first two plaintext-ciphertext pairs but not the secret key. You intercept the ciphertext consisting of IV_1, IV_2, C_1, and C_2. Your goal is to modify the ciphertext so that when Bob decrypts it, he obtains (P_1', P_2') instead of the original (P_1, P_2). How should you alter IV_1, IV_2, C_1, and C_2 to achieve this?

$$\begin{aligned} \mathrm{IV}_1' &= \mathrm{IV}_1 \oplus P_1 \oplus P_1' \\ \mathrm{IV}_2' &= \mathrm{IV}_2 \oplus P_2 \oplus P_2' \\ C_1' &= C_1 \oplus P_2 \oplus P_2' \\ C_2' &= C_2 \end{aligned}$$